ANVIL COMPOUNDS

RESEARCH USE ONLY

● RESEARCH GRADE · 99%+ PURITY · USA-BASED

RESEARCH REFERENCE & SAFETY DOCUMENTATION

KLOW

Four-peptide research blend · Research-grade reference material

TYPE	CONSTITUENTS	FORM	USE
Multi-component blend	**4**	**Lyophilized**	**In-vitro research only**

01 COMPOSITION

What Is In This Blend

KLOW is a four-component lyophilized blend combining a melanocortin-derived anti-inflammatory tripeptide, a copper tripeptide complex, and two tissue-repair research peptides. Each constituent is documented separately below.

Constituent	Research area	Class
KPV	Anti-inflammatory research	MELANOCORTIN-DERIVED TRIPEPTIDE
GHK-Cu	ECM / skin-biology research	COPPER(II) TRIPEPTIDE COMPLEX
BPC-157	Tissue-repair research	SYNTHETIC PENTADECAPEPTIDE (15 AA)
TB-500	Tissue-repair research	THYMOSIN B4 FRAGMENT

How to read this document. Each constituent is documented individually in Section 02, because that is how the research literature exists — these compounds have been studied separately, not as this combination. Section 03 addresses what that means for interpreting blend data. The Safety Data Sheet covers the mixture, drawing on constituent and class information.

Quantitation. Confirm the per-vial mass of each constituent against the lot certificate of analysis before preparing solutions. Blend ratios are set by the supplier and are not standardized across the industry, so results are not directly comparable between vendors or lots unless the composition matches.

Constituents — Part 1

Findings below derive from animal and cell-culture systems and pertain to each compound studied **individually**. They are not established effects in humans, and they are not findings about this blend.

KPV — Anti-inflammatory research

Class	Melanocortin-derived tripeptide
Origin	α-MSH(11–13) C-terminal fragment
Sequence	Lys-Pro-Val
Formula / MW	$C_{16}H_{30}N_4O_4$ / 342.44

- Inhibits NF-κB nuclear translocation via an intracellular, receptor-independent route
- Reduces MAPK cascade activation
- Suppresses TNF-α, IL-1β and IL-6 output in epithelial models
- Enters intestinal cells via the PepT1 transporter

Evidence: Kannengiesser et al. (2007) reported significant anti-inflammatory effects in two murine colitis models, partially independent of MC1R. Notably, KPV retains anti-inflammatory activity without melanocortin receptor binding or the pigmentary effects of parent α-MSH. Preclinical only; no approved human indication.

GHK-Cu — ECM / skin-biology research

Class	Copper(II) tripeptide complex
CAS (complex / free)	89030-95-5 / 49557-75-7
MW	≈403.9 (complex) / 340.38 (free)
Composition	Cu(II)–Gly-His-Lys, 1:1

- Collagen (I, III) and glycosaminoglycan synthesis in dermal fibroblast culture
- MMP modulation — “balanced” matrix remodeling
- Lysyl oxidase (copper-dependent) crosslinking activity
- Downregulation of TNF-α / IL-6; antioxidant enzyme modulation

Evidence: Among the most replicated tripeptides in skin/ECM research since the 1980s. Best human-relevant evidence is topical; injectable/systemic use is poorly supported and has been flagged by FDA over limited human safety data.

Constituents — Part 2

BPC-157 — Tissue-repair research

Class	Synthetic pentadecapeptide (15 aa)
CAS	137525-51-0
Formula / MW	$C_{62}H_{98}N_{16}O_{22}$ / 1419.56
Sequence	GEPPPGKPADDAGLV

- Angiogenesis reported via the VEGFR2–Akt–eNOS axis
- FAK–paxillin and ERK1/2 signaling in cell-migration models
- Growth-hormone receptor upregulation in tendon fibroblast cultures
- Reduced pro-inflammatory cytokines in injury models

Evidence: Extensive preclinical literature (a 2025 HSS Journal systematic review screened 544 articles, including 36 — 35 preclinical, 1 clinical). No large controlled human efficacy trials. WADA-prohibited.

TB-500 — Tissue-repair research

Class	Thymosin β4 fragment
Parent CAS / MW	77591-33-4 / ≈4921
Active domain	LKKTETQ
Fragment identity	Confirm against supplier documentation

- High-affinity 1:1 G-actin sequestration (reported Kd ≈ 0.5–0.7 μM)
- Keratinocyte and fibroblast migration in wound models
- Endothelial migration and tubule formation (LKKTET motif implicated)
- NF-κB-associated inflammatory modulation

Evidence: Most primary evidence concerns full-length thymosin β4 rather than the fragment. The legitimate clinical programme is RGN-259 (ophthalmic, Phase 3, no approved product). No human RCT supports systemic musculoskeletal use. WADA-prohibited.

The Blend Question — Read This Carefully

This is the single most important section in a blend document, and it is the one most often omitted. The constituents of this product have published preclinical literature. **The combination does not.**

What this means in practice

PER-CONSTITUENT EVIDENCE	COMBINATION STUDIES	INTERACTION DATA	STANDARDIZED RATIOS
Exists (preclinical)	**Essentially none**	**Not characterized**	**None industry-wide**

- **Evidence does not simply add up.** Preclinical findings for KPV, GHK-Cu, BPC-157, TB-500 were generated with each compound administered alone, at defined doses in defined models. Combining the compounds does not combine their evidence bases.
- **Interactions are uncharacterized.** Additive, synergistic and antagonistic effects between these constituents have not been systematically studied. Absence of reported interactions is not evidence of their absence.
- **Ratios are not standardized.** Blend composition varies between suppliers and lots, so published work — and results from other vendors' blends — are not directly transferable.
- **Attribution is confounded.** An observed effect cannot be attributed to a single constituent without appropriate single-compound controls.
- **Design implication.** Where mechanism matters, single-compound arms are necessary. A blend suits combination-effect questions, not characterization of any individual constituent.

> **Appropriate characterization:** a multi-component research preparation whose constituents each carry preclinical literature, and whose combined behaviour is uncharacterized. Claims describing the blend as delivering the summed or amplified effects of its parts are not supported by published evidence.

Regulatory & status notes

US FDA	No constituent of this blend is an approved drug. The blend is not approved for any use.
WADA	Certain constituents appear on the World Anti-Doping Agency Prohibited List — verify per constituent against the current list.
Supply basis	Research reagent for in-vitro laboratory use only; not for human or veterinary administration.

Selected references

1. Vasireddi N, et al. Emerging Use of BPC-157 in Orthopaedic Sports Medicine: A Systematic Review. HSS Journal. 2025.
2. Thymosin Beta-4 and TB-500 in Tissue Healing, Regeneration, and Musculoskeletal Repair: A Scoping Review. Applied Sciences. 2026;16(12):6202.
3. Maquart FX, et al. Stimulation of collagen synthesis by GHK-Cu^{2+} in fibroblast cultures. FEBS Lett. 1988. PMID 3169264.
4. Kannengiesser K, et al. Melanocortin-derived tripeptide KPV has anti-inflammatory potential in murine models of inflammatory bowel disease. PMID 18092346.

Safety Data Sheet

Prepared per GHS Rev. 8 / OSHA HazCom 2012
(29 CFR 1910.1200) · 16-section format

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	KLOW
Type	Multi-component blend (see §3)
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use
Supplier	
Emergency	

2 Hazard Identification

Not classified based on currently available data for the mixture. **No toxicological data exist for this combination as a mixture** — hazard information below is derived from the individual constituents and their chemical classes. Absence of classification is not a determination of absence of hazard. **Precautionary:** P261, P264, P280, **P273 (avoid release to environment)**, P501.

3 Composition

Multi-component blend. Constituent identities:

KPV	Melanocortin-derived tripeptide
GHK-Cu	Copper(II) tripeptide complex
BPC-157	Synthetic pentadecapeptide (15 aa)
TB-500	Thymosin β4 fragment

Confirm the per-vial mass of each constituent against the lot certificate of analysis. Residual synthesis reagents, solvents and counter-ions may be present. **Contains chelated copper(II)** via the GHK-Cu component.

4 First-Aid Measures

Eyes: rinse with water for several minutes. **Skin:** wash with soap and water; remove contaminated clothing. **Inhalation:** move to fresh air. **Ingestion:** rinse mouth; do not induce vomiting unless directed. Ingestion of copper compounds may cause gastrointestinal irritation. Treat symptomatically; no specific antidote known.

5 Fire-Fighting

Media: CO_2, dry chemical, foam, water spray. Combustion may produce CO, CO_2, nitrogen oxides and copper-containing particulate. Use SCBA and full protective gear.

6 Accidental Release

Avoid dust formation; ventilate; use PPE per §8. Do not allow entry to drains or waterways — copper compounds may be harmful to aquatic life. Collect in a closed labeled container for disposal per §13.

7 Handling & Storage

Handle as a mixture of bioactive substances of incompletely characterized toxicity; minimize all routes of exposure. Weigh and reconstitute in a fume hood or ventilated enclosure. Store the lyophilized solid tightly closed, protected from light, moisture and heat; −20 °C long-term, 2–8 °C short-term. Protect from light — the copper-peptide component is light- and oxidation-sensitive. Aliquot reconstituted solutions; store frozen; avoid repeated freeze-thaw. Observe the COA retest date.

8 Exposure Controls / PPE

No established OEL for any constituent; control exposure to ALARA. General copper-dust/fume limits may inform handling. Local exhaust or fume hood for the dry powder. Nitrile gloves (double-glove for neat powder); ANSI Z87.1 eye protection; lab coat; N100/P100 respirator per 29 CFR 1910.134 where controls are insufficient. Eyewash/safety shower per ANSI Z358.1.

9 Physical & Chemical Properties

State	Lyophilized powder
Appearance	Blue-tinted to blue-violet powder (copper-peptide component)
Odor	Odorless
Solubility	Soluble in water
MW	Not applicable — mixture
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable as a dry solid under recommended storage; protect from light. Avoid heat, moisture, UV, open flame. Incompatible with strong oxidizers, strong acids and bases, strong reducing agents, and strong chelators that could displace copper from the GHK-Cu component. Decomposition: CO, CO_2, NOx, copper oxides. No hazardous polymerization.

11 Toxicological Information

No toxicological data exist for this combination as a mixture. Constituent data are individually limited: no LD_{50}/LC_{50} values are available for any component from authoritative databases. Skin/eye: treat as potentially irritating. Copper compounds can be irritating. Sensitization, mutagenicity, carcinogenicity, reproductive toxicity, STOT and aspiration: no authoritative substance-specific data; not classified based on available data; hazards cannot be excluded. Potential interactive or additive effects between constituents have not been characterized. Avoid exposure if pregnant, nursing, or planning pregnancy.

12 Ecological Information

The copper-containing constituent can be toxic to aquatic organisms. As a precaution treat the mixture as potentially hazardous to the aquatic environment. No substance-specific ecotoxicity, persistence or bioaccumulation data identified for the mixture or its constituents. Avoid release to surface water, soil, drains or sewers. A qualified reviewer should confirm whether GHS aquatic-toxicity classification applies.

13 Disposal

Dispose of contents and container per applicable local, state and federal regulations; copper content may affect waste classification. US: 40 CFR Parts 261–270 (RCRA). Do not dispose into sewers or waterways. Use a licensed waste disposal company.

14 Transport Information

Not expected to be regulated as dangerous goods under DOT (49 CFR), IATA or IMDG based on current classification, **but confirm** given copper content and potential marine-pollutant considerations. UN number: not applicable. The shipper must independently verify requirements.

15 Regulatory Information

US TSCA: constituents may qualify for the R&D exemption (40 CFR 720.36) depending on conditions of use. OSHA HazCom 2012 aligned with GHS Rev. 8. EU REACH: supplied for Scientific Research and Development use; the importer/user must verify any exemption. No constituent of this blend is an FDA-approved drug. Not a legal determination of regulatory status.

16 Other Information

Revision [DATE] · Version 1.0. Independent review by a qualified chemical-safety professional is recommended prior to use — note the copper-specific classification items above. Because this is a mixture, hazard determination should address both constituent and combination effects. Sources: PubChem; peer-reviewed literature; OSHA HazCom 2012; GHS Rev. 8. Provided without warranty. For scientific R&D use only.